

Mixtures & Alligations

COMPLETE LEARNING GUIDE

What You'll Learn

Building on the basic two-ingredient alligation rule, placement tests frequently extend mixture problems into more intricate territory: repeatedly withdrawing and replacing part of a mixture with a pure ingredient (the classic milk-and-water dilution problem), mixing three or more ingredients at once, working backward from a final ratio to find how much of an ingredient must be added, and applying the alligation cross to non-price quantities like exam marks or population averages. The core alligation rule never changes — ratio = (higher value " mean) : (mean " lower value) — but recognizing which value plays which role in a disguised word problem is the real skill being tested. This sub-module works through repeated dilution, multi-ingredient mixing, reverse alligation, and non-price alligation applications with complete worked numbers, extending the foundational rule into every commonly tested variant.

1. The Alligation Cross: Quick Recap

To mix two ingredients of different values to hit a target mean value, the ratio in which they must be combined is (higher value " mean) : (mean " lower value) — this comes directly from balancing the 'excess' and 'deficit' around the mean.

- Mix milk (₹50/L) with water (₹10/L) to get a mixture worth ₹40/L. Ratio milk:water = (40 " 10):(50 " 40) = 30:10 = 3:1.

2. Repeated Dilution: Replacing Part of a Mixture With Pure Liquid

When a fixed quantity is withdrawn from a vessel and replaced with a pure second liquid, and this process repeats n times, the amount of the ORIGINAL liquid remaining follows: final quantity = $P \times (1 - x/P)^n$, where P is the total volume and x is the amount withdrawn (and replaced) each time. This is not a linear reduction — each successive withdrawal removes a smaller absolute amount of the original liquid because it's more diluted.

- A vessel contains 40 L of pure milk. 4 L is withdrawn and replaced with water, and this is repeated 3 times in total. Milk remaining = $40 \times (1 - 4/40)^3 = 40 \times (0.9)^3 = 40 \times 0.729 = 29.16$ L.

3. Single Withdrawal-and-Replacement as a Special Case

A single withdrawal-and-replacement is just the $n=1$ case of the repeated dilution formula, and can be verified directly by simple subtraction — useful as a sanity check before trusting the formula on a multi-step version.

- A vessel has 20 L of pure milk. 4 L is withdrawn (leaving 16 L of milk) and replaced with 4 L of water. Direct check: 16 L milk remains. Formula check: $20 \times (1 - 4/20)^1 = 20 \times 0.8 = 16$ L.

4. Mixing Three or More Ingredients

When quantities of three or more ingredients are already known (as a ratio), the resulting mixture's average value is simply the weighted average: $(\sum V_i \cdot \text{value}_i) \div \sum V_i$. When quantities are EQUAL, this collapses to the plain arithmetic mean of the values.

- Three teas costing \$50, \$60, and \$65 per kg are mixed in the ratio 2:3:5. Average price = $(2 \times 50 + 3 \times 60 + 5 \times 65) / (2+3+5) = (100+180+325)/10 = 605/10 = \60.50 per kg.

5. Reverse Alligation: Finding the Quantity to Change a Given Ratio

Some problems give a mixture's current ratio and total volume, then ask how much of one ingredient must be ADDED to reach a new target ratio. Since adding pure ingredient changes only that ingredient's quantity (the other stays fixed), set up a direct equation using the unchanged term.

- A 24 L mixture of milk and water is in the ratio 5:1 (so 20 L milk, 4 L water). How much water must be added to make the ratio 5:3? Milk stays at 20 L. $20/(\text{new water}) = 5/3 \Rightarrow \text{new water} = 12$ L. Water to add = $12 - 4 = 8$ L.

6. Alligation Applied to Non-Price Averages

The alligation rule works identically for any weighted-average scenario, not just prices — exam marks, population growth rates, or speeds split between two sub-groups can all be alligated the same way once you identify which quantity is the 'higher value,' which is the 'lower value,' and which is the overall mean.

- A class's overall average score is 68. Students who passed averaged 75; students who failed averaged 40. Find the percentage of students who passed. Ratio passed:failed = $(75 - 68) : (68 - 40) = 7 : 28 = 1 : 4$. So passed fraction = $4/5 = 80\%$.

7. Adding Pure Ingredient to an Existing Mixture

Adding a pure ingredient to an existing mixture increases only that ingredient's amount while the other ingredient's quantity stays fixed — this is a direct application of the reverse-alligation logic in Section 5, but framed as 'add milk' rather than 'add water,' and is common enough to warrant its own recognition.

- A 40 L mixture contains milk and water in the ratio 3:1 (30 L milk, 10 L water). How much pure milk must be added to change the ratio to 4:1? Water stays at 10 L. $(30+x)/10 = 4/1 \Rightarrow 30+x = 40 \Rightarrow x = 10$ L of milk must be added.

Common Pitfalls

- Setting up the alligation cross with the subtraction order reversed — it's always (higher " mean) : (mean " lower), not the other way around.
- In repeated-dilution problems, subtracting the withdrawn fraction n times (linear approximation) instead of raising $(1 - x/P)$ to the power n .
- Forgetting that withdrawing and replacing with water keeps the TOTAL volume constant — only the milk quantity decreases.
- In reverse-alligation problems, changing both terms of the ratio proportionally instead of recognizing that adding a pure ingredient only changes ONE term.
- Assuming a three-or-more-ingredient mixture's average is always the simple arithmetic mean, forgetting this is only true when quantities are equal.
- Working with ratio numbers directly as if they were actual quantities, instead of first converting the ratio into real liters/kg using the given total.
- Applying alligation to a scenario that requires a straightforward weighted average, adding unnecessary complexity instead of computing $(2\frac{1}{2} \text{ G} \cdot w_{\text{blue}}) / (2\frac{1}{2} \text{ G} + 7\frac{1}{2} \text{ F})$.

Key Takeaways

- Alligation ratio = (higher value " mean) : (mean " lower value).
- Repeated dilution: final quantity of original liquid = $P \times (1 - x/P)^n$, where x is withdrawn and replaced each of n times.
- Weighted mixture value = $(2\frac{1}{2} \text{ V} \cdot \frac{1}{2} \text{ F} + 7\frac{1}{2} \text{ G} \cdot \frac{1}{2} \text{ F}) \div (2\frac{1}{2} \text{ V} + 7\frac{1}{2} \text{ G})$ — this equals the simple average only when all quantities are equal.
- To change a mixture's ratio by adding a pure ingredient, only that ingredient's term changes — set up the equation using the fixed term.
- Total volume stays constant when a portion is withdrawn and replaced with another liquid of the same volume.
- Alligation applies to any weighted-average scenario (marks, rates, speeds), not just prices — identify the higher value, lower value, and mean first.

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GeeksforGeeks — Alligation or Mixture Aptitude Questions

<https://www.geeksforgeeks.org/aptitude/aptitude-alligation-or-mixture/>

IndiaBix — Alligation or Mixture Aptitude Questions

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